

Assignment-01

My Student Id is : 0112410532

Q.1 $n = (32 \% 5) + 5$
 $\therefore n = 7$

```
#include <stdio.h>
int main() {
    int i, j, k, num;
    int n = 7;
    for(i = 1; i <= n; i++) {
        for(j = n; j > i; j--) {
            printf(" ");
        }
        num = 1;
        for(k = 1; k <= i; k++) {
            printf("%d", num);
            num += 2;
            if (num > n) {
                num = 1;
            }
        }
        printf("\n");
    }
    return 0;
}
```

File

Sta



"E:\New folder\Untitled1.exe" X



```
-----1
-----13
-----135
----1357
--13571
_135713
1357135
```

Process returned 0 (0x0) execution time: 0.000 s
Press any key to continue.

Q.2

my student Id last 4 digit is 0532

But I count only 532 for decimal calculation.

$$\text{int } a = \text{532} / 7; \quad = 76$$

$$\text{int } b = 532 \% 7; \quad = 0$$

$$\text{float } c = (\text{float}) 532 / 7; \quad = 76.000000$$

$$\text{float } d = (\text{float}) (532 / 7); \quad = 76.000000$$

$$\text{int } e = (a - b) < 0 \&\& e; \quad = 0$$

So the answer

$$\begin{aligned} a &= 76 \\ b &= 0 \\ c &= 76.000000 \\ d &= 76.000000 \\ e &= 0 \end{aligned}$$

Q:3

a)

My student ID last 2 digits is 32

Now, $num = 32 \% 5;$

That means, $num = 2$

if we enter 2 as a Input on this program

Then,

consider, case 2:

where, $sum = 0$

$sum + 2 = j - 2;$

$j = 3$

$j++;$

$j = 1$

break;

and then, $sum = 1$

$j = 4$

$j = 1$

The output is :

for i for j for sum
1 4 1

Q:3

b)

Rewrite the code Q.3a "switch" to ~~an~~ "if-else"

```
#include <stdio.h>
```

```
int main () {
```

```
    int num;
```

```
    int sum = 0, i = 1, j = 3;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &num);
```

```
    if (num == 1) {
```

```
        sum += (i + j);
```

```
        i++;
```

```
        sum += j - 2;
```

```
        j++;
```

```
    }
```

```
    else if (num == 2) {
```

```
        sum += j + - 2;
```

```
        j++;
```

```
    }
```

```
    else if (num == 3) {
```

```
        sum  
        sum = (i + j) * i;
```

```
        i++;
```

```
        j--;
```

```
        sum -= (i + j);
```

```
        j--;
```

```
    }
```

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```
else if (num == 4) {
```

```
    sum -= (i + j);
```

```
    j--;
```

```
}
```

```
printf("c%d d%d e%d", i, j, sum);
```

```
return 0;
```

```
}
```

```

1      #include <stdio.h>
2
3      int main() {
4
5
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28

```

"E:\New folder\Untitled4.exe" X + v

```

2
1 4 1
Process returned 0 (0x0)   execution
Press any key to continue.
|

```

Q.4

my student Id last Three digit is 532

$$n = 532 \% 66$$

that means, $n = 4$;

now,

$$n = 4, i = 0, \text{Value_final} = 0.$$

Loop, $i = n - 4$ $0 \leq n$
 $= 4 - 4$ $0 \leq 4$ - True
 $= 0$,

$$\begin{aligned} \text{Value_final} &= \text{Value_final} + (i * i) \\ &= 0 + (0 * 0) \\ &= \underline{0} \end{aligned}$$

$++i$ - that means $i = 0 + 1 = 1$ // Statement

$i++$ - that means $i = 1 + 1 = 2$ // Increment
else

end the first round,

$$\begin{aligned} \text{Value_final} &= \underline{0} \\ i &= \underline{2} \end{aligned}$$

Now,

$$i \leq n$$

$$2 \leq 4 - \text{True}$$

$$\begin{aligned} \text{Value_final} &= \text{Value_final} + (i * i) \\ &= 0 + (2 * 2) \\ &= \underline{4} \end{aligned}$$

$$++i; \text{ — that means } i = 2 + 1 = 3 \text{ // Statement}$$

$$i++ \text{ — that means } i = 3 + 1 = \underline{4} \text{ // Increment}$$

end 2nd round,

$$\begin{aligned} \text{Value_final} &= \underline{4} \\ i &= \underline{4} \end{aligned}$$

Now,

$$i \leq n$$

$$4 \leq 4 - \text{True}$$

$$\begin{aligned} \text{value_final} &= \text{value_final} + (i * i) \\ &= 4 + (4 * 4) \\ &= \underline{20} \end{aligned}$$

$$++i; \text{ — that means } i = 4 + 1 = 5 \text{ // Statement}$$

$$i++ \text{ — that mean } i = 5 + 1 = \underline{6} \text{ // Increment}$$

end the 3rd round,

$$\begin{aligned} \text{value_final} &= \underline{20} \\ i &= \underline{6} \end{aligned}$$

}

Now,

$$i \leq n$$

$$6 \leq 4 \text{ false ;}$$

~~program~~ loop stop

last value of "value_final" = 20

last value of "i" = 6.

Q.5

my student Id last Three digit is 532

$$n = 532 \% 66$$

$$= 4$$

